Slide 1 — Challenges and the automation opportunity

“Today, primer skin inspection involves considerable manual effort, from detecting and measuring defects to recording the results. Access to some areas is difficult, and inspection capacity can become a bottleneck.

The chart illustrates another challenge: different operators can obtain different measurements for the same defect. This makes consistent assessment more difficult.

The opportunity is to use visual inspection to identify areas that need further investigation, then guide the NDT sensor to those specific regions.

The overall objective is more consistent inspection, less manual effort and better traceability. Our contribution focuses on automatically planning the robot movements needed for this targeted inspection.”

Slide 2 — Scope of development

“Once the defect locations are available, the next question is: how do we inspect them efficiently?

A curved panel may contain more than a hundred potential regions. Programming every visit and every local scan individually creates significant preparation work.

Our development covers three areas.

First, mission planning: choosing an efficient order to visit the regions and reducing travel between them.

Second, automatic motion generation: creating a scan pattern adapted to each region, then generating robot movements that respect the process and equipment constraints.

Third, validation: checking the planned movements and the complete inspection sequence before execution.

Together, these capabilities aim to reduce programming effort and unnecessary movement, freeing up inspection capacity.”

Slide 3 — Solution proposal

“This slide shows how the approach works.

We provide three types of input: the defect locations, the scan settings, and the robot and panel models with the relevant constraints.

The scan pattern is configurable. We can change its area, shape and swathe width to suit the region and the sensor.

The system then determines the visiting order and generates the scan paths and robot movements. During planning, it checks for collisions and problematic robot configurations, while seeking an efficient path.

Next, we verify the generated program in ROBOGUIDE. This allows us to identify execution issues before running it on the physical robot.

After validation, the program is executed on the robot.

The important point is that a new panel or a different set of defect locations becomes a new planning input, reducing the need to program each movement manually.”

Slide 4 — Demonstration

“Here, you can see the planning demonstration alongside the physical robot setup.

There are three things to look for in the demonstration.

First, the sequence of visits: how the robot moves between the selected inspection regions.

Second, the local coverage: how the back-and-forth scanning pattern covers each region.

Third, the robot motion: how those scan paths translate into movements that respect the robot and process constraints.

The scan pattern can be adjusted when the region changes, and the robot motion can then be regenerated.

This demonstrates the core capability: automatically turning inspection regions into a planned robot inspection sequence.”

[Pause to show the demonstration.]

Slide 5 — Benefits and value added

“The value comes from three practical improvements.

Adaptable scan paths mean less manual programming and faster preparation when the panel or defect locations change.

An optimized task sequence reduces unnecessary travel. Combined with targeted scanning, this creates the opportunity for shorter inspection cycles.

Motion checks and program validation help identify problems earlier, reducing physical trial-and-error and issues during execution.

There is also a wider opportunity: reusing the planning capability across projects. Different applications can use the planner option best suited to their requirements, while sharing common interfaces and validation methods. This reduces repeated development effort.

We will quantify the gains through preparation time, inspection cycle time and physical trial effort.

The message is simple: **less programming for each new part, more efficient inspection, and a planning capability we can reuse.**”